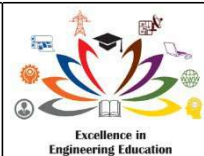




Pune Vidyarthi Griha's
COLLEGE OF ENGINEERING, TECHNOLOGY
AND MANAGEMENT, PUNE-09



Pune Vidyarthi Griha's College of Engineering, Technology and Management

Department of Artificial Intelligence and Data Science

Subject Name & Code:
417532(B): Business Intelligence

Student Name: Shreya Yadav

Roll No.: 22028088

Batch : B3

Date of Submission: 20-04-2026

ASSIGNMENT:- 1

AIM:

Load Data from Multiple Sources into a Target System

PROBLEM STATEMENT :

Import the legacy data from different sources such as Excel , Sql Server, Oracle etc and load in the target system

THEORY:

This assignment demonstrates the fundamentals of **Business Intelligence (BI)** and data integration using the **ETL (Extract, Transform, Load)** process. In real-world scenarios, organizations store data across multiple platforms such as Excel files, SQL Server databases, and Oracle systems. To enable effective decision-making, this distributed data must be centralized, cleaned, and made ready for analysis using BI tools like Power BI.

Let's briefly understand the components involved:

1. Microsoft Excel:

Excel is one of the most commonly used spreadsheet tools for storing structured data in tabular form. It allows users to perform basic data entry, sorting, filtering, and analysis using formulas and charts. In many organizations, Excel serves as a quick and flexible way to record and maintain business data like sales records, customer lists, or financial reports. In the ETL process, Excel is often the **source system** from which data is extracted.

2. SQL Server:

Microsoft SQL Server is a **relational database management system (RDBMS)** designed to handle large-scale, structured data efficiently. It stores data in tables with relationships and supports complex querying using Structured Query Language (SQL). SQL Server is often used in enterprise systems for transactional data such as inventory, HR, finance, and customer interactions. It also supports advanced operations like stored procedures, indexing, and data security. In BI, SQL Server acts as a robust data source for structured and relational datasets.

3. Oracle Database:

Oracle is another powerful **RDBMS**, widely used in large enterprises and known for its scalability, performance, and advanced security features. It supports complex transactions and massive datasets, and is often used in mission-critical applications such as ERP, CRM, and banking systems. Oracle databases also allow

querying using SQL and PL/SQL. Like SQL Server, Oracle is a frequent source in ETL pipelines, especially when integrating legacy or enterprise-grade data.

4.Power BI:

Power BI is a **Business Intelligence and Data Visualization tool** developed by Microsoft. It enables users to connect to a wide range of data sources, transform data using the Power Query Editor, and create interactive dashboards and reports. Power BI plays the role of the **target system** in the ETL process. Once data is loaded into Power BI, users can build visualizations such as bar charts, maps, and KPIs to gain insights and support strategic business decisions.

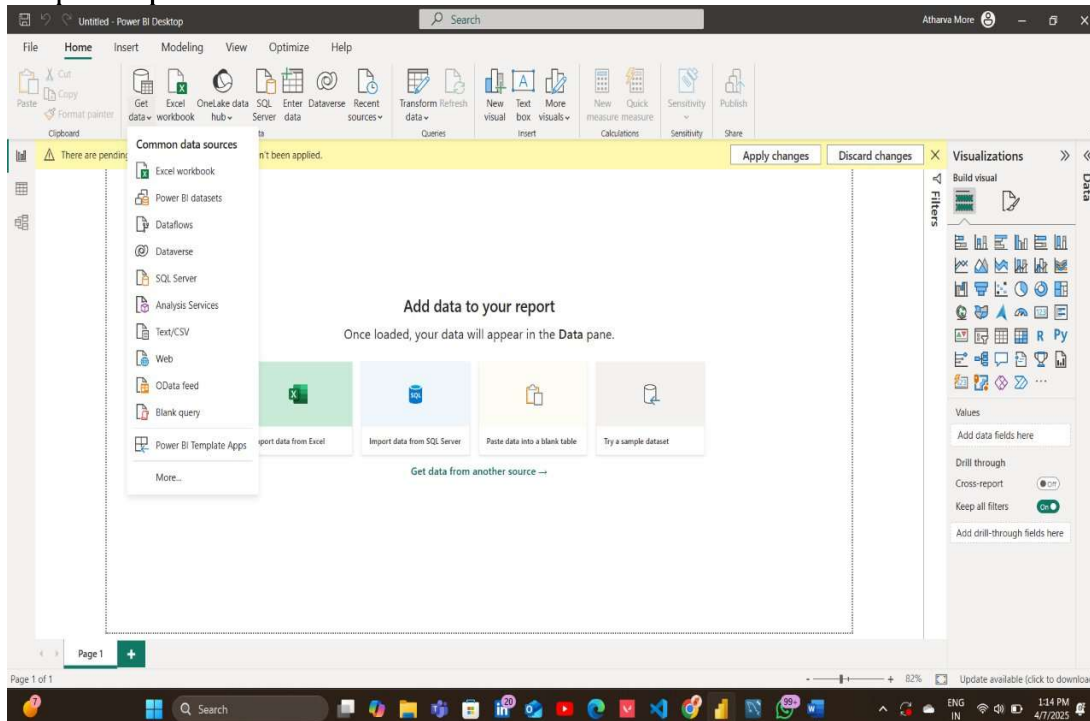
Overall Workflow in BI Context:

- **Extract** data from sources like Excel, SQL Server, and Oracle.
- **Transform** it by cleaning, reshaping, or aggregating using Power Query.
- **Load** it into Power BI for analysis and reporting.

This integrated approach ensures the data is accurate, consistent, and ready to support data-driven decision-making.

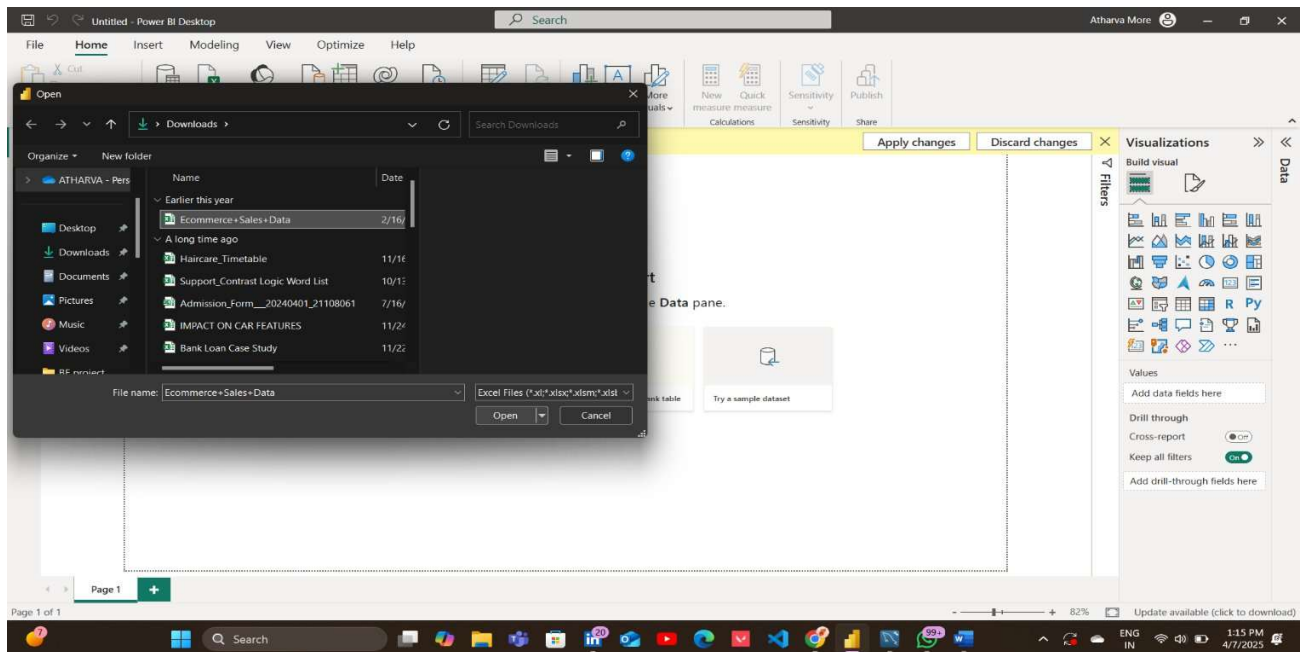
PROCEDURE :

Step 1: Open Power BI

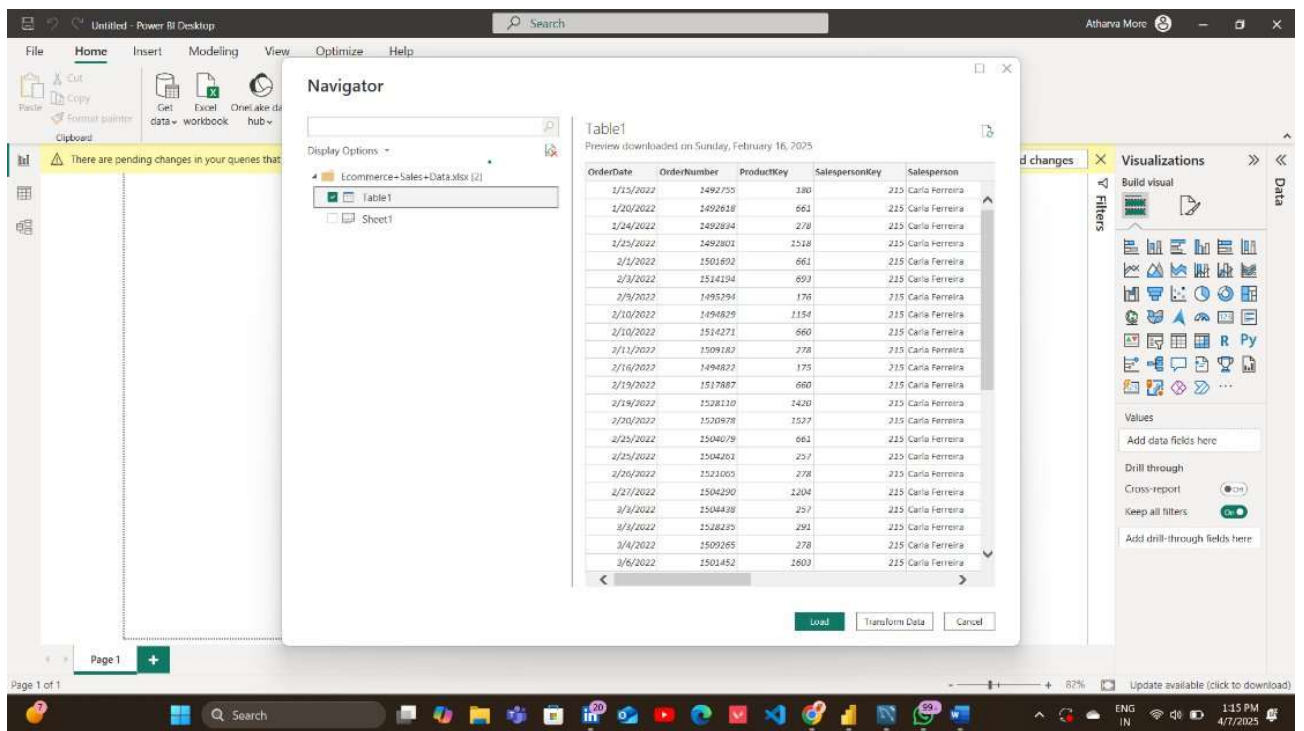


Step 2 : Click on Get data following list will be displayed → select Excel

Step 3: Select required file and click on Open, Navigator screen appears



Step 4: Power query editor appears



Untitled - Power BI Desktop

File Home Insert Modeling View Optimize Help

Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Refresh Preview Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type Date Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Test Analytics Vision Azure Machine Learning

This preview may be up to 50 days old. Refresh

Table.TransformColumnTypes(Table1_Table,{{"OrderDate", type date}, {"OrderNumber", Int64.Type}, {"ProductKey",

	OrderDate	OrderNumber	ProductKey	SalespersonKey	Salesperson	Supervisor
1	1/15/2022	1492755	180	215	Carla Ferreira	Diego Araujo
2	1/20/2022	1492618	661	215	Carla Ferreira	Diego Araujo
3	1/24/2022	1492834	278	215	Carla Ferreira	Diego Araujo
4	1/25/2022	1493801	1518	215	Carla Ferreira	Diego Araujo
5	2/1/2022	1501692	662	215	Carla Ferreira	Diego Araujo
6	2/3/2022	1514194	693	215	Carla Ferreira	Diego Araujo
7	2/9/2022	1495294	176	215	Carla Ferreira	Diego Araujo
8	2/10/2022	1494829	1154	215	Carla Ferreira	Diego Araujo
9	2/10/2022	1514271	660	215	Carla Ferreira	Diego Araujo
10	2/11/2022	1509182	278	215	Carla Ferreira	Diego Araujo
11	2/16/2022	1494822	175	215	Carla Ferreira	Diego Araujo
12	2/19/2022	1517887	660	215	Carla Ferreira	Diego Araujo
13	2/19/2022	1528110	1420	215	Carla Ferreira	Diego Araujo
14	2/20/2022	1520978	1527	215	Carla Ferreira	Diego Araujo
15	2/25/2022	1504079	661	215	Carla Ferreira	Diego Araujo
16	2/25/2022	1504261	257	215	Carla Ferreira	Diego Araujo
17	2/26/2022	1521065	278	215	Carla Ferreira	Diego Araujo
18	2/27/2022	1504290	1204	215	Carla Ferreira	Diego Araujo

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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File Home Insert Modeling View Optimize Help

Power Query Editor

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Close & Apply New Recent Enter Data Data source settings Manage Parameters Refresh Preview Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type Date Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Test Analytics Vision Azure Machine Learning

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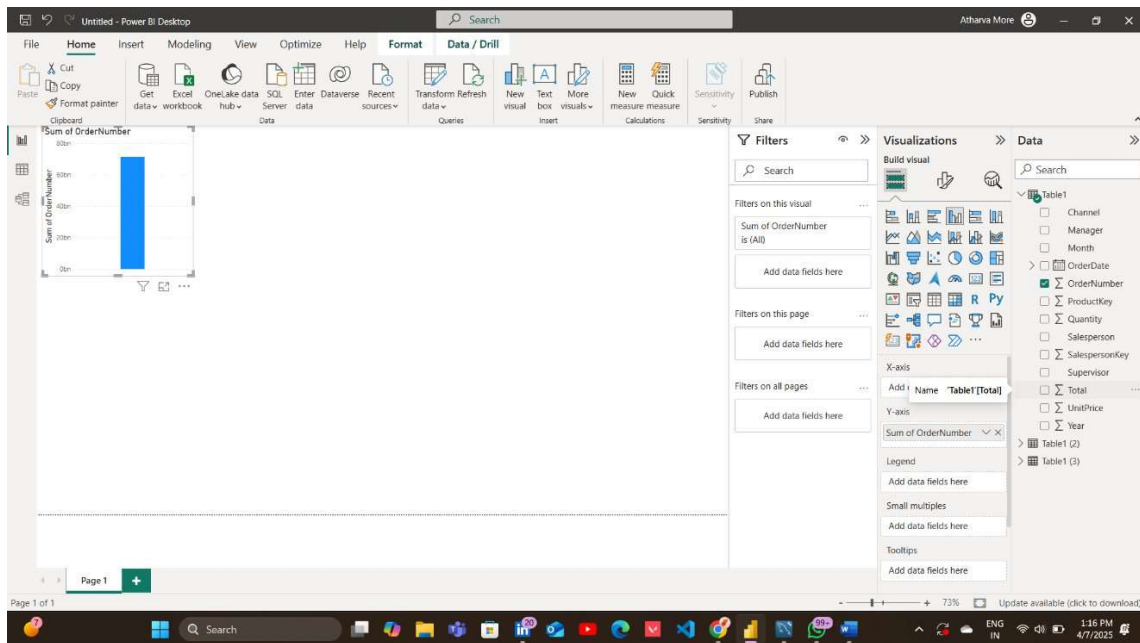
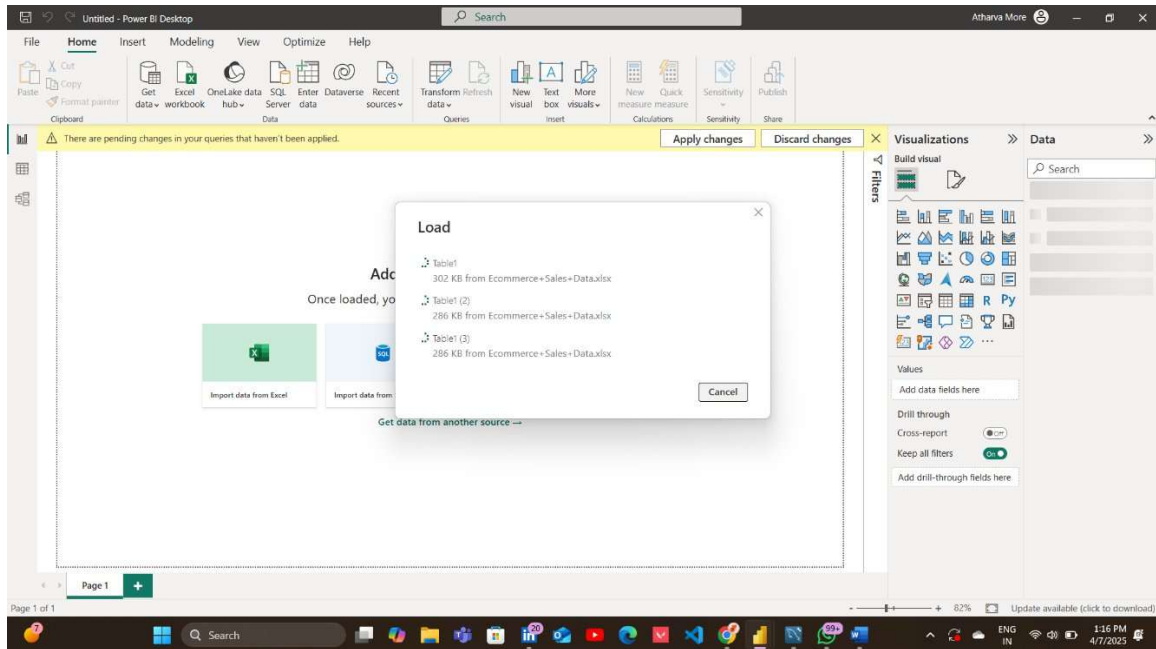
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Step 5: Load data and start visualisation



OD Data Feed

Step 6: Paste url as <http://services.odata.org/V3/Northwind/Northwind.svc/>
Click on ok

The screenshot displays the Power BI Desktop interface. In the foreground, the 'OData feed' dialog box is open, showing the 'Basic' tab with the URL <http://services.odata.org/V3/Northwind/Northwind.svc/> entered. The background shows a bar chart titled 'Sum of OrderNumber' and the 'Data' pane on the right, which lists fields like Channel, Manager, Month, OrderDate, OrderNumber, ProductKey, Quantity, Salesperson, SalespersonKey, Supervisor, Total, UnitPrice, and Year. Below the OData feed dialog, the 'Navigator' window is open, showing a list of tables on the left and a preview of the 'Order_Details' table on the right. The 'Order_Details' table preview shows columns: OrderID, ProductID, UnitPrice, Quantity, Discount, and Order. The table contains 20 rows of data.

OrderID	ProductID	UnitPrice	Quantity	Discount	Order
10248	11	14	12	0	Record
10248	42	9.8	10	0	Record
10248	72	34.8	5	0	Record
10249	14	18.6	9	0	Record
10249	51	42.4	40	0	Record
10250	41	7.7	10	0	Record
10250	51	42.4	35	0.150000006	Record
10250	65	16.8	15	0.150000006	Record
10251	22	16.8	6	0.050000001	Record
10251	57	15.6	15	0.050000001	Record
10251	63	16.8	20	0	Record
10252	20	64.8	40	0.050000001	Record
10252	33	2	25	0.050000001	Record
10252	60	27.2	40	0	Record
10253	31	10	20	0	Record
10253	39	14.4	42	0	Record
10253	49	16	40	0	Record
10254	24	3.6	15	0.150000006	Record
10254	55	19.2	21	0.150000006	Record
10254	74	8	21	0	Record
10255	2	15.2	20	0	Record
10255	16	13.9	35	0	Record
10255	36	15.2	25	0	Record

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File Home Insert Modeling View Optimize Help Format Data / Drill

Query Settings

Properties

Name: Order_Details

Applied Steps

Source

Navigation

OrderID	ProductID	UnitPrice	Quantity	Discount	Order
1	10248	11	14	12	0 Record
2	10248	42	9.8	10	0 Record
3	10248	72	34.8	5	0 Record
4	10249	14	18.6	9	0 Record
5	10249	51	42.4	40	0 Record
6	10250	42	7.7	10	0 Record
7	10250	51	42.4	35	0.150000006 Record
8	10250	65	16.8	15	0.150000006 Record
9	10251	22	16.8	6	0.050000001 Record
10	10251	57	15.6	15	0.050000001 Record
11	10251	65	16.8	20	0 Record
12	10252	20	64.8	40	0.050000001 Record
13	10252	33	2	25	0.050000001 Record
14	10252	60	27.2	40	0 Record
15	10253	31	10	20	0 Record
16	10253	39	14.4	42	0 Record
17	10253	49	16	40	0 Record
18	10254	24	3.6	15	0.150000006 Record
19	10254	55	19.2	21	0.150000006 Record

7 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Load

Order_Details Evaluating...

Cancel

Query Settings

Properties

Name: Order_Details

Applied Steps

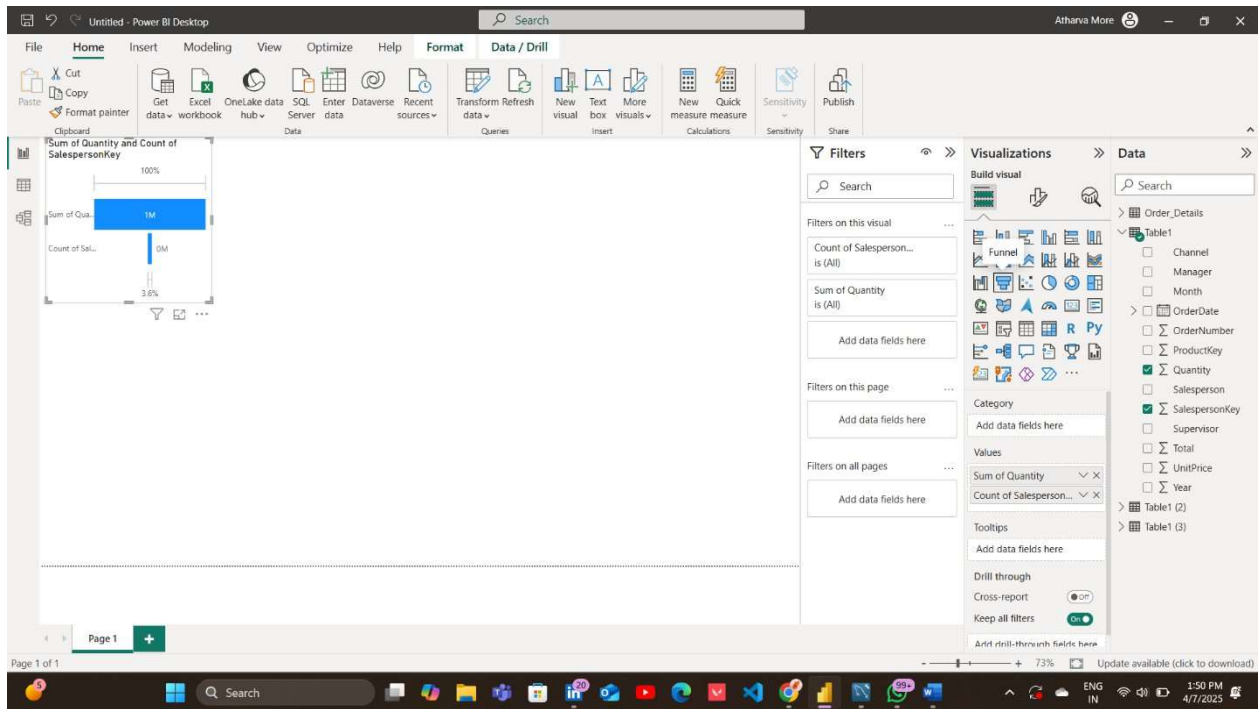
Source

Navigation

OrderID	ProductID	UnitPrice	Quantity	Discount	Order
1	10248	11	14	12	0 Record
2	10248	42	9.8	10	0 Record
3	10248	72	34.8	5	0 Record
4	10249	14	18.6	9	0 Record
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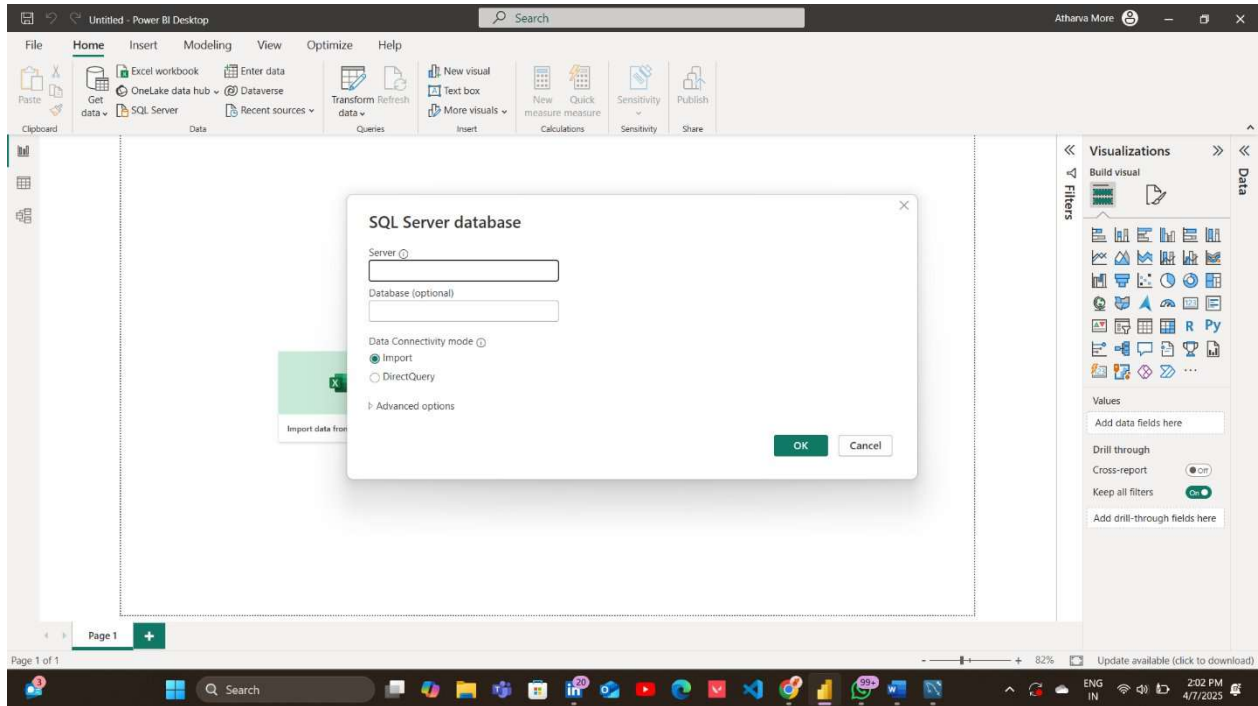
7 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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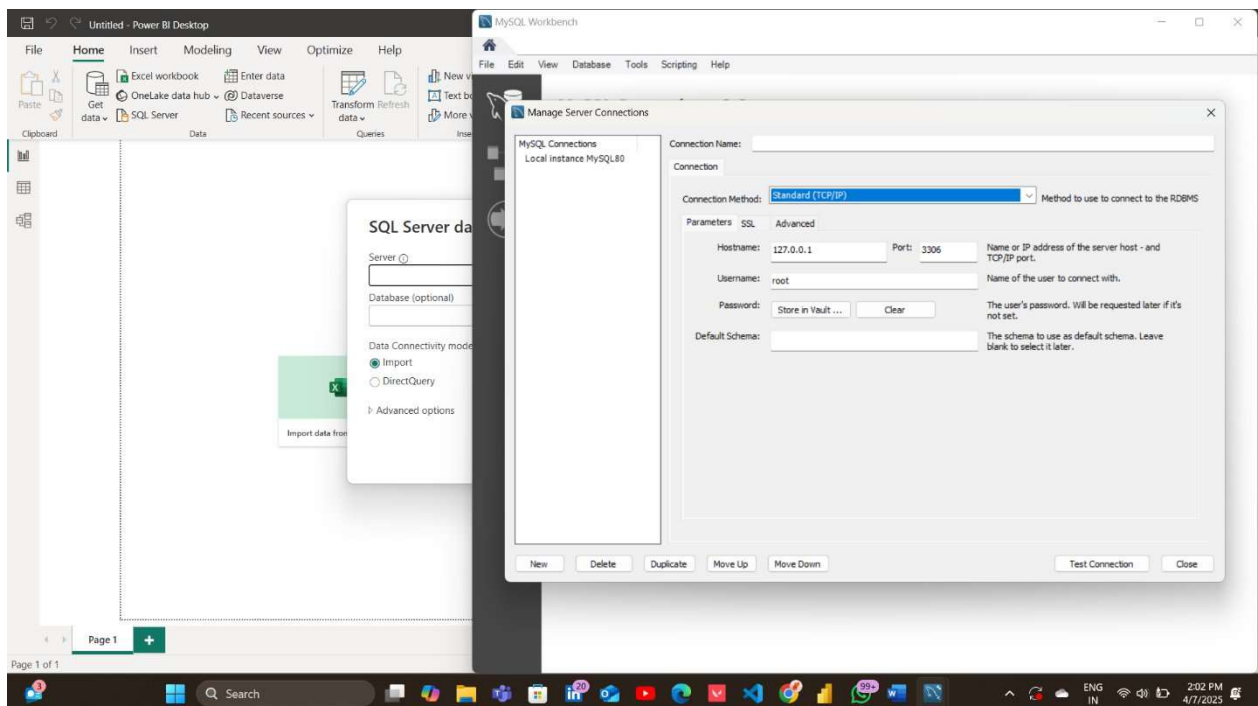


SQL SERVER

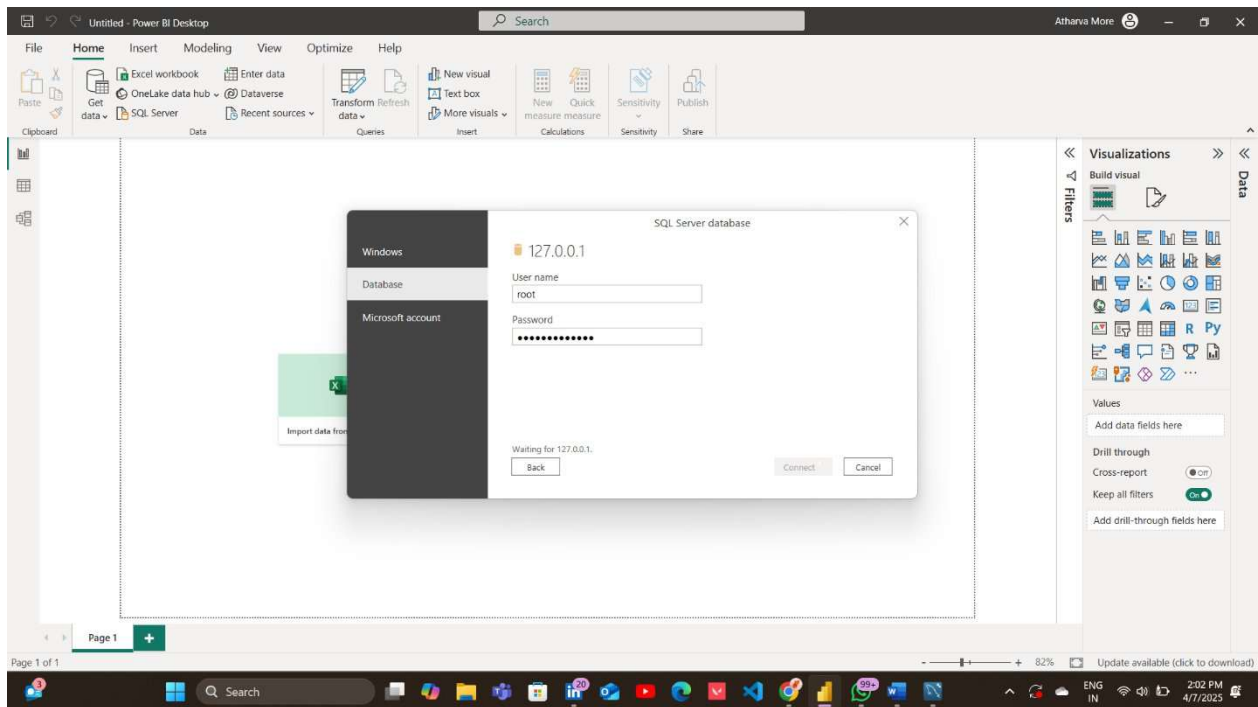
Step 1: Select SQL server from the Get Data



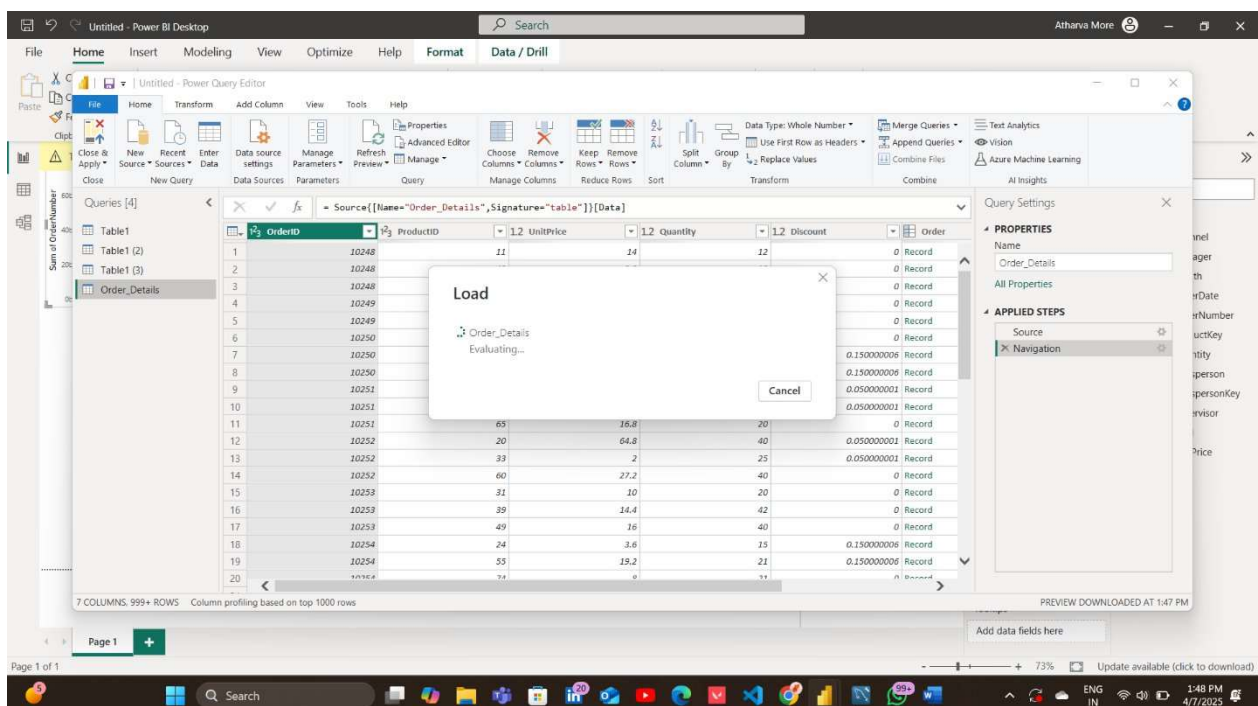
Step 2: Select the root or port number from the SQL server manage connections



Step 3: Enter User root and Password and establish the connection



Step 4: Import the Database from the SQL workbench



CONCLUSION:

Integrating data from multiple sources such as Excel, SQL Server, and Oracle plays a vital role in creating a centralized and consistent data environment. Through the ETL process, data is effectively extracted, transformed, and loaded into a target system, enabling efficient analysis and reporting. This enhances decision-making and supports better business insights.